

Week 8 updates

Attempted fixes

- Log-normal time sampling to over-emphasize intermediate times - extremely unstable sampling when trained on small number of epochs, might work with a lot more
- Translation of x_1 by +10 resulted in the model outputting 0
- Train from $x_0 \sim \mathcal{N}(0, 1)$ but clamp to $[-1, 1]$ - mostly reduces exploding outliers but results in low-diversity output

Variable-sized clouds

- Can't find bug in zero padding
- Potential fix: apply curriculum learning approach. Sort training data such that average jet size per batch decreases over time

Potential fix for exploding output

- Return $x^L - x_0$ instead of x^L - when the model performs poorly and returns an identity output, we return 0 as opposed to a linearly exploding output

Rotation invariance

Apply approach of Song et al. 2023 to permute and rotate particles in $x_0 \sim \mathcal{N}$ to align with $x_1 \sim p_{\text{data}}$ in every (x_0, x_1) training pair

Algorithm 3 A variant of iterative closest point (ICP) algorithm.

- 1: **Input:** a point cloud $\mathbf{z} \in \mathbb{R}^{3 \times N}$ and its reference point cloud $\mathbf{y} \in \mathbb{R}^{3 \times N}$.
 - 2: **while** τ has not converged **do**
 - 3: Obtain permutation matrix $\mathbf{\Pi} = \underset{\mathbf{\Pi}}{\operatorname{argmin}} \|\mathbf{\Pi}(\mathbf{R}\mathbf{z})^\top - \mathbf{y}^\top\|_2$ with Jonker-Volgenant algorithm [50]
 - 4: Obtain rotation matrix $\mathbf{R} = \underset{\mathbf{R}}{\operatorname{argmin}} \|\mathbf{R}(\mathbf{\Pi}\mathbf{z})^\top - \mathbf{y}^\top\|_2$ with Kabsch algorithm [51]
 - 5: $\tau = \|\mathbf{\Pi}(\mathbf{R}\mathbf{z}^\top)^\top - \mathbf{y}^\top\|_2$
 - 6: **end while**
 - 7: **return** $\mathbf{\Pi}, \mathbf{R}$
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Boost invariance

- Song et al. 2023 achieve translation invariance by projecting x_0 to zero COM space during training, and projecting x_t to zero COM at every step of the ODE solver
- We have the target jet's 4-momenta, so we can boost the noisy particles x_0 from their own reference frame to the reference frame of the jet at initialization, and similarly boost

Coordinate space

Approach of Spinner et al. 2024:

- Polar coordinates are more physically meaningful than Cartesian coordinates, but working in Cartesian coordinates is better for equivariance
- The network works entirely in Cartesian coordinates, and its outputs are projected to polar coordinates
- Loss is calculated in terms of polar coordinates
- Should mitigate the zero-collapse issue